

Quick Notes

Extension of ABC Theorem (9.26.2026):

The ABC theorem applies specifically to a self-adjoint Hamiltonian on Euclidean configuration space (an operator $H = -\Delta + V(x)$ on $L^2(\mathbb{R}^n)$), with V satisfying an appropriate dilation-analyticity condition. The following is a paper wherein Kelvin extends this theorem to a more general setting. It's terse, and I don't think it has a use for us, but I put it here in case.

<https://arxiv.org/abs/1003.2538>

Issue with Complex ϕ

$$u_{k,l}(r) \rightarrow \hat{h}_l^+(kr) \sim e^{ikr}.$$

For $k = k_r - i\gamma$ ($k_r, \gamma > 0$),

$$u_{k,l}(r) \rightarrow e^{ik_r r} e^{\gamma r}.$$

Take $r \rightarrow re^\theta$:

$$u_{k,l}(re^\theta) \rightarrow e^{ik_r re^\theta} e^{\gamma re^\theta}.$$

Continuing to $\theta = \theta_r + i\theta_i$:

$$u_{k,l}(re^\theta) \rightarrow e^{ik_r re^{\theta_r} e^{i\theta_i}} e^{\gamma re^{\theta_r} e^{i\theta_i}}$$

$$u_{k,l}(re^\theta) \rightarrow e^{i\tilde{\theta} k_r r (\cos \theta_i + i \sin \theta_i)} e^{\gamma \tilde{\theta} r (\cos \theta_i + i \sin \theta_i)},$$

where $\tilde{\theta} \rightarrow e^{\theta_r}$.

$$u_{k,l}(re^\theta) = e^{i\tilde{\theta} r (k_r \cos \theta_i + \gamma \sin \theta_i)} e^{\tilde{\theta} r (\gamma \cos \theta_i - k_r \sin \theta_i)}.$$

This converges as long as $\gamma \cos \theta_i - k_r \sin \theta_i < 0 \rightarrow \theta_i > |\arg(E)|/2$, as already established by the ABC theorem. We further require that $V(re^\theta)$ remains well behaved. For a Gaussian $V(r) = V_0 \exp(-(r - r_0)^2/w^2)$. For simplicity, assume $V(r) = V_0 e^{-r^2/w^2}$. Then,

$$\begin{aligned} V(re^\theta) &= V_0 \exp(-r^2 e^{2\theta}/w^2) = V_0 \exp(-r^2 e^{2\theta_r} e^{2i\theta_i}/w^2) \\ &= V_0 \exp[-\tilde{\theta}^2 r^2/w^2 (\cos(2\theta_i) + i \sin(2\theta_i))] = V_0 e^{-\tilde{\theta}^2 r^2/w^2 \cos(2\theta_i)} e^{-i\tilde{\theta}^2 r^2/w^2 \sin(2\theta_i)}. \end{aligned}$$

First, we see that when $\theta_i \geq \pi/4$, $-\tilde{\theta}^2 r^2/w^2 \cos(2\theta_i) > 0$, so the potential diverges. Thus, we need $\theta_i < \pi/4$ to remain well-behaved. $|\arg(E)|/2 < \theta_i < \pi/4$ [Gaussian]. Second, we see that the radial coordinate is stretched as $r \sim r\tilde{\theta}$. This has the following effect: the DVR mesh (n, L) (where n is the number of mesh points and L is the system length) is fixed upon creation. When $\tilde{\theta}$ is small, we're effectively reduce our sampling from $r \in [0, L]$ to $r \in [0, L\tilde{\theta} \ll L]$. We've refined our mesh width by a factor $\tilde{\theta}$, but we cut off information past $L\tilde{\theta}$. If the potential is still meaningful beyond that point, we'll pick up a domain truncation error (IR error). Further, if $\tilde{\theta}$ is very large, then we've effectively increased our sampling region from $r \in [0, L]$ to $r \in [0, L\tilde{\theta} \gg L]$, but we've effectively coarsened our mesh. If the relevant scale for the potential is l_{phys} , and we effectively coarsen our mesh

to $\Delta = \tilde{\theta}w \gg w > l_{\text{phys}}$, then we're undersampling, and we can't represent the relevant physics (UV error).

This is an issue for finding plateaus: scanning over a complex θ grid looking for a plateau in observables will be contaminated by whatever FV errors are accumulating underneath it. So, a real, physical plateau can look un-converged due strictly to FV effects, and a seemingly-converged plot might actually be physically unstable. I'd have to adjust the mesh width every time I adjusted θ_r if I wanted to account for this exactly (similarly to how I do when scanning over L), but I can't train EC on training vectors coming from different DVR systems (unless FVEC?), so I have to pick a mesh width and stick with it. I can scan for a plateau in the θ grid, then nudge n/L a little (make my mesh finer) to see if it remains converged at the most demanding point (near the edge). If it does, then it should be a sufficiently real plateau.